

Empowering Sustainable Living with ORES

A Cloud-Native Approach to Software-Defined Home Energy Network

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LF Energy ORES Working Group



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What is ORES?

- **ORES: Open Renewable Energy Systems**
- **A working group under LF Energy**
- Builds open architecture and standards for decentralized renewable energy systems
- Focused on **cloud-native software + hardware abstraction + interoperability**
- Use cases: microgrids, home grid energy networks, community DER coordination, started currently **pilot deployments** for **houses in California**

Why ORES?

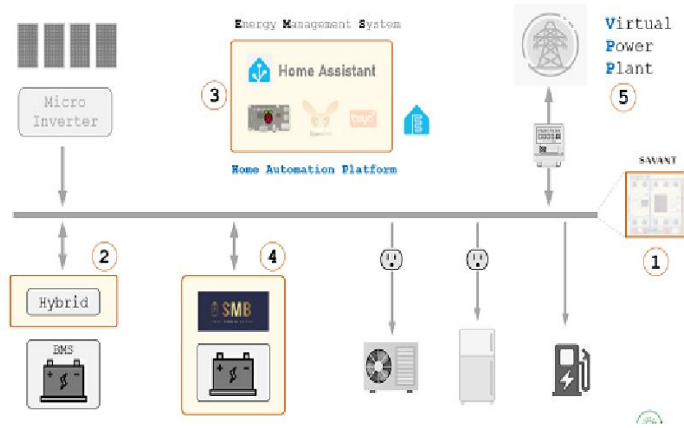
Problem: Traditional home energy systems are siloed, inflexible, and hardware-bound: **device & deployment cost** .

Need: **open source, standardized, Innovation**, interoperable, and intelligent systems to integrate DERs, storage, and smart appliances.

Vision: Empower households with plug-and-play, software-defined energy autonomy, and **a total solution with cloud native & AI.**



Open Source & Standardized System



Solar Panel + Battery/BMS + PCS +
Software Defined EMS

Dimensions to address:

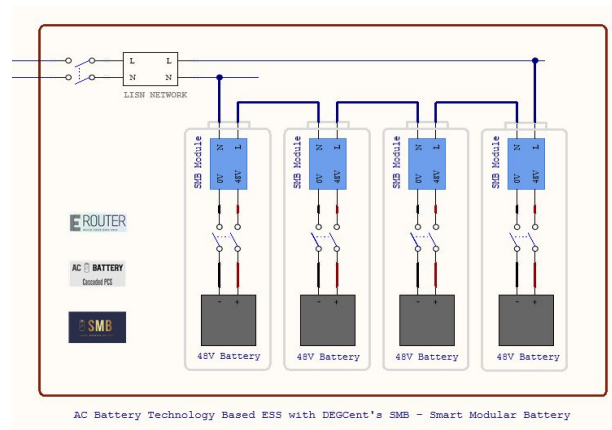
- 1) Hardware device level
- 2) Compatible communication and interconnection
- 3) Software Defined EMS
- 4) Deployment, safety monitoring and operation maintenance.
- 5) **Cloud Native & AI**: Battery on-line, off-line safety and performance evaluation and analysis, etc.

Pilot Implementation & Innovation

“AC Battery” - Smart Modular Battery low-voltage Cascaded PCS Technology



- DIY support, LV, Drop & Play
- Reduce deployment cost



- Increase Safety
- Reduce equipment cost

Cloud-Native Home Energy Architecture

Requirements & Expectation

**Deployment, monitoring,
maintenance, safety**



Software-Defined EMS (Energy Management System)

- Built on Kubernetes-native architecture
- RESTful APIs, declarative energy states
- Flexible plug-in model for energy devices (inverters, batteries, EVs, HVAC, etc.)

To the Edge (Hardware): Modular smart devices with firmware agents, "**Energy IoT**"

To the Cloud: Real-time telemetry, distributed orchestration, resilience at scale

Introducing GAIFARE

Generative AI for Autonomous Renewable Energy

- A new initiative incubated under ORES proposed
- Uses generative AI to power intelligent agents across home devices
- Enables **autonomous coordination of energy production, storage, and consumption, and safety**
- Context-aware learning: weather, pricing, usage patterns, peer negotiation
- Builds on the software-defined foundation of ORES to enable the *next-gen autonomous grid*

Call to Action

- Join us in building the **open source future of energy**
- Contribute to ORES projects (software, architecture, AI models, hardware protocols)
- Collaborate on GAIFARE and other AI-powered workstreams
- Be part of a community shaping **cloud-native, intelligent, and sustainable energy systems**



Connect!



ORES Github



ORES Website

Thank you!